European Journal of Cardio-Thoracic Surgery 2024, 65(4), ezae082
https://doi.org/10.1093/ejcts/ezae082 Advance Access publication 28 March 2024

ORIGINAL ARTICLE

THORACIC

Cite this article as: Leviel F, Fourdrain A, Delatre F, De Dominicis F, Lefebvre T, Bar S *et al*. Serratus anterior plane block alone, paravertebral block alone and their combination in video-assisted thoracoscopic surgery: the THORACOSOPIC double-blind, randomized trial . Eur J Cardiothorac Surg 2024; doi:10.1093/ejcts/ezae082.

Serratus anterior plane block alone, paravertebral block alone and their combination in video-assisted thoracoscopic surgery: the THORACOSOPIC double-blind, randomized trial

Florent Leviel [a], Alex Fourdrain [b], Florian Delatre[a], Florence De Dominicis [c], Thomas Lefebvre[a], Stéphane Bar [a], Hamza Yahia Alshatri [d], Emmanuel Lorne [e], Olivier Georges[c], Pascal Berna [f], Hervé Dupont [a], Jonathan Meynier [g] and Osama Abou-Arab [a,*]

[a] Department of Anesthesia and Critical Care, Amiens Hospital University, Amiens, France
[b] Department of Thoracic Surgery, Marseille University Hospital, Marseille, France
[c] Department of Thoracic Surgery, Amiens Hospital University, Amiens, France
[d] Faculty of Medicine, Department of Anesthesia and Critical Care, University of Jeddah, Jeddah, Saudi Arabia
[e] Department of Anesthesia and Critical Care Medicine, Millénaire Clinic, Montpellier, France
[f] Department of Thoracic Surgery, Victor Pauchet Clinic, Amiens, France
[g] Department of Biostatistics, Amiens Hospital University, Amiens, France

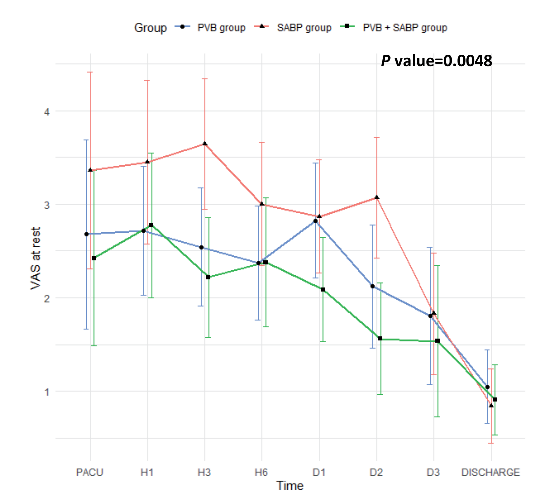
* Corresponding author. Department of Anesthesia and Critical Care, Amiens Hospital University, 1 rue du Professeur Christian Cabrol, F-80054 Amiens, France. Tel: +33-322-087-836. e-mail: osama.abouarab@gmail.com (O. Abou-Arab).

Received 16 November 2023; received in revised form 20 February 2024; accepted 27 February 2024

Serratus anterior plane block alone, paravertebral block alone, and their combination in video-assisted thoracoscopic surgery: the THORACOSOPIC double-blind, randomized trial

Summary

The benefit of a combination of SAVP +BVP to decrease postoperative pain in VATS. A double-blind, single center, randomized trial comparing SABP alone, BVP alone and SABP+BVP in 156 VATS patients. VAS did not differ between groups at anesthesia wake.
Pain score was significantly lower in SABP+BVP groups during hospital stay.



Legend: **PACU:** post awake care unit; **PVB:** paravertebral block; **SAPB:** serratus anterior plane block; **VATS:** video assisted thoracic surgery; **VAS:** visual analogue scale.

Presented at the French Society of Anesthesia and Critical Care meeting (SFAR) at Paris, September 2023.

Abstract

OBJECTIVES: Serratus anterior plane block (SAPB) and paravertebral block (PVB) are well known to reduce pain levels after video-assisted thoracoscopic surgery (VATS). However, the relative efficacies of each block and a combination of the 2 have not been fully characterized. The objective of the present study was to assess the efficacy of PVB alone, SAPB alone and the combination of PVB and SAPB with regard to the occurrence and intensity of pain after VATS.

METHODS: We conducted the THORACOSOPIC single-centre, double-blind, randomized trial in adult patients due to undergo elective VATS lung resection. The participants were randomized to PVB only, SAPB only and PVB + SAPB groups. The primary end-point was pain on coughing on admission to the postanaesthesia care unit. The secondary end-points were postoperative pain at rest and on coughing at other time points and the cumulative opioid consumption. Pain was scored on a visual analogue scale.

RESULTS: One-hundred and fifty-six patients (52 in each group) were included. On admission to the postanaesthesia care unit, the 3 groups did not differ significantly with regard to the pain on coughing: the visual analogue scale score was 3 (0–6), 4 (0–8) and 2 (0–6) in the PVB, SAPB and PVB + SAPB groups, respectively ($P = 0.204$). During postoperative care, the overall pain score was significantly lower in the SABP + PVP group at rest and on cough.

CONCLUSIONS: The combination of SABP + PVB could be beneficial for pain management in VATS in comparison to SABP or PVB alone.

Keywords: Video-assisted thoracoscopic surgery • Pain • Regional anesthesia • Paravertebral block • Serratus anterior plane block

ABBREVIATIONS

ESRA	European Society of Regional Anaesthesia
PACU	Postanaesthesia care unit
PVB	Paravertebral block
SAPB	Serratus anterior plane block
VAS	Visual analogue scale
VATS	Video-assisted thoracoscopic surgery

INTRODUCTION

In the era of enhanced recovery after surgery, minimally invasive surgery has become a standard of care in many fields. In thoracic surgery, video-assisted thoracoscopic surgery (VATS) has been in the European Society of Thoracic Surgery's 2019 guidelines [1]. The guidelines also state that pain management (typically based on loco-regional anaesthesia) is mandatory for successful enhanced recovery after surgery. There is a large body of literature data on regional anaesthesia in thoracic surgery, including paravertebral block (PVB) and serratus anterior plane block (SAPB). Recently, the European Society of Regional Anaesthesia (ESRA) issued guidelines on the indications and application of PVB and SAPB [2].

According to the ESRA guidelines, PVB is indicated as the first-line technique for patients undergoing VATS [2]. PVB targets the intercostal spinal nerve contained in the thoracic paravertebral space. Hence, PVB allows the blockade of thoracic nerve root [3]. With the transition from thoracotomy to VATS, PVB has supplanted conventional epidural anaesthesia. Indeed, PVB is not inferior in terms of acute postoperative pain levels and opioid consumption, and the technique's benefits are no longer subject to debate [4, 5]. However, PVB may provide incomplete pain relief with an insufficient anaesthesia coverage—notably in the anterior part of the chest wall [6]. Alternatively, SAPB might avoid the limitations of PVB and might produce better anaesthesia of the anterior chest wall. SABP targets the lateral branches of the intercostal nerves with local anaesthesia injection deep or superficial to the serratus anterior muscle. The emergence of ultrasound and the ease of implementation have facilitated the spread of SAPB. In fact, SAPB is more superficial than PVB, and the serratus anterior plane is easier to identify using ultrasound. Pain scores are lower with SAPB than with no block [7, 8]. In contrast, SAPB alone does not confer sufficient anaesthesia of the posterior chest wall.

We hypothesized that the combination of SAPB and PVB could provide complete anaesthesia of the anterior, lateral and posterior chest walls. Hence, the objective of the present study was to assess the efficacy of PVB alone, SAPB alone and the combination of PVB and SAPB with regard to postoperative pain in patients having undergone VATS.

METHODS

Ethics

This single-centre, double-blind, randomized trial was conducted at Amiens Hospital University (Amiens, France). The study was approved by an independent ethics committee (*Comité de Protection des Personnes Ouest II*, Angers, France) on 15 October 2019 (identifier: PI2019_843_0051) and was performed in line with the French legislation on clinical research [9]. Participants received written information on the study and gave their written, informed consent prior to inclusion. The trial was registered in the European Clinical Trials Database (identifier: EudraCT 2019-002858-23) and in the ClinicalTrials.gov registry (identifier: NCT04222010). The study's results were reported in compliance with the Consolidated Standards of Reporting Trials statement [10].

Population study

The study population comprised adult patients scheduled for video-assisted sublobar lung resection (segmentectomy or wedge) or lobar lung resection for suspected or confirmed non-small-cell bronchopulmonary cancer. The exclusion criteria were age under 18 years, legal guardianship, pregnancy, refusal of regional anaesthesia, a personal medical history of morphine-treated chronic pain and inability to assess pain.

Intervention

All regional anaesthesia blocks were performed after the induction of the general anaesthesia and after the patient had been moved from the prone position to the lateral position. All blocks were guided by ultrasound, using a high-frequency linear probe. Strict asepsis was required prior to initiation of the block, with sterile gloves, a sterile probe cover, a surgical mask and a head covering [11]. The skin was prepared with an alcohol-based chlorhexidine gluconate solution. PVB was performed at the T5–T6 level. The needle was inserted in an in-plane approach up to the paravertebral space, i.e. between the superior costotransverse ligament and parietal pleura [12]. The total volume injected was 20 ml. Deep SAPB was performed between the 6th and 7th intercostal spaces. In an in-plane approach, the needle was inserted into the mid-axillary line between the serratus anterior muscle and the rib [12]. The total volume injected was 40 ml [12].

Ropivacaine (0.2%) or placebo (0.9% saline serum) were injected on a blind basis, depending on the group allocation. Hence, all patients received both PVB (real or sham) and SAPB (real or sham). In the SAPB group, patients received SAPB with 40 ml of ropivacaine and sham PVB with 20 ml of saline serum. In the PVB group, patients received sham SAPB with 40 ml of saline serum and PVB with 20 ml of ropivacaine. In the PVB + SAPB group, patients received SAPB with 40 ml of ropivacaine and PVB with 20 ml of ropivacaine.

To ensure blinding, three 20 ml syringes were delivered to the operating theatre for each patient. Two syringes were labelled 'serratus anterior plane block' and 1 syringe was labelled 'paravertebral block'. In agreement with the central pharmacy, the syringes were prepared by a nurse anaesthesiologist not involved in the patient's care.

Anaesthesia management

The anaesthesia procedure was standardized according to national guidelines and local protocols. Anaesthesia was induced with a bolus of propofol (3–4 mg kg^{-1}), remifentanil (target controlled infusion, with a plasma concentration from 5 to 9 ng ml^{-1}) and a bolus of neuromuscular blockade agent (cisatracurarium or atracurium, at the physician's discretion). A single bolus of ketamine 0.15 mg kg^{-1} was administered at the physician's discretion. Before induction, preoxygenation was performed in all patients. We used a double-lumen tracheal tube for intubation. Anaesthesia was maintained with sevoflurane, controlled infusion of remifentanil (target controlled plasma concentration: 2–5 ng ml^{-1}) and intermittent boluses of neuromuscular blocking agent. The depth of anaesthesia was monitored as the bispectral index in all patients, with a target value of between 40 and 60. A train-of-four neuromuscular monitor watch on the ulnar nerve was used in all patients; the neuromuscular response had to be 0 out of 4. Mechanical ventilation was set with a tidal volume of between 6 and 8 ml/kg of ideal body weight, a positive end-expiratory pressure of 5–10 cm H_20, an inspired oxygen fraction that gave an oxygen saturation value of over 92% and a respiratory rate that gave an end-tidal CO_2 value between 35 and 40 mmHg. During open lung ventilation, the tidal volume was set to 3 ml/kg of ideal body weight. The neuromuscular blockade was reversed with neostigmine and atropine in all patients, in order to achieve a T4/T1 response ratio >0.9 [13].

Intraoperative pain management included dexamethasone (8 mg), paracetamol (1 g), oxycodone (0.15 mg/kg) and nefopam (20 mg). Ketoprofen 100 mg was administered if not contraindicated (a history of gastric ulcers, cardiovascular disease or chronic antiplatelet drug use). After surgery, the patient was admitted to our postanaesthesia care unit (PACU). The postoperative pain management protocol included paracetamol (1 g every 6 h), nefopam (20 mg every 6 h), ketoprofen (100 mg every 12 h) and tramadol (50 mg every 6 h). When visual analogue scale (VAS) for pain score was over 3, opioid titration was proceeded by delivering 1 mg of oxycodone every 7 min until VAS < 3.

Surgical management

VATS was performed in line with our institution's standard procedures. The surgical indication was approved for all patients in a multidisciplinary team meeting comprising an oncologist, a pulmonologist, a radiologist and a thoracic surgeon. Fitness for surgery (nutritional status, smoking cessation, anaemia management, a rehabilitation programme when indicated and lung function testing) was assessed for all patients. The VATS procedure was standardized, using a modified three-port anterior approach and all ports in the 7th intercostal space [14]. All patients received a single chest drain at the end of the procedure. A digital drainage system was used for all patients, and the drain was removed in the absence of air leaks and residual pleural effusion. The patient was discharged from hospital after drain removal; there were no outpatients with chest drains.

Definition and assessment of end-points

The primary end-point was pain on coughing on admission to the PACU. Pain was rated on a VAS ranging from 0 (lowest) to 10 (highest).

The secondary end-points were pain at rest and on coughing at the following time points: PACU admission, 1, 2, 3 and 6 h and 1, 2 and 3 days after PACU admission, and on hospital discharge. The other end-points were the cumulative morphine consumption during 72 h (mg), the duration of chest drainage (days), the length of hospital stay (days) and the presence of pneumonia (a composite criterion with a pulmonary infiltrate on a plain X-ray and at least 2 of the following: (i) fever, (ii) new or continuing postoperative inflammatory syndrome and (iii) clinical symptoms (focal findings on auscultation, purulent sputum and/or dyspnoea).

Neuropathic pain was defined as a Douleur Neuropathique 4 (DN4) questionnaire graded at 3 or over at month 6 after surgery [15, 16]. Douleur Neuropathique 4 value at month 6 was reported.

Randomization and blinding

Prior to VATS, the study participants were randomly assigned to 1 of 3 groups. The randomization code was generated with Clinsight® software. To ensure that the groups were balanced, a random block size of 6 was applied. Eligible patients were randomized 1:1:1 without stratification to PVB and sham SAPB (forming the PVB group), SAPB and sham PVB (forming the SAPB group) or PVB and SAPB (forming the PVB + SAPB group). The randomization list was masked to study participants, investigators and study monitors until the database was locked.

Treatment dispensing and concealment was managed by the hospital's central pharmacy in accordance with the randomization plan.

Sample size

With a Cohen's d of 0.5, a power of 80% and a bilateral alpha risk of 5%, a power analysis showed that 156 patients (52 in each group) would be required to assess the primary end-point. We did not expect any loss to follow-up.

Statistical analysis

Continuous variables were presented as the median (interquartile range) and the categorial variables as frequency (percentage). For the continuous variables, the 3 groups were compared with a Kruskal–Wallis test. For the categorical variables, Chi-squared test were used or Fisher's exact test if a cell was at 5 or less in the contingency table.

Intergroup differences in the change over time in the VAS pain score (at rest and on coughing) were probed in linear mixed models (random-intercept model and restricted maximum likelihood for the maximization). The threshold for statistical significance was set to $P < 0.05$. Statistical analyses were performed using SAS software (version 9.4, SAS Institute, Inc., Cary, NC, USA) and R software (version 4.2.3). 'lme4' and 'lmerTest' packages were used for the linear mixed models. 'CompareGroups' package was used assess the variable distribution and to proceed in groups comparisons.

RESULTS

The study population

Between January 2020 and July 2022, 1297 patients were screened for inclusion (Fig. 1) and 156 of the latter were randomized (52 in each group; Table 1). Regional anaesthesia failed in 1 patient in the PVB group and in 1 in the SAPB group. All patients were included in an intention-to-treat-analysis.

Primary end-point

On admission to the PACU, the groups did not differ significantly with regard to the level of pain on coughing: the VAS pain score was 3 (0–6), 4 (0–8) and 2 (0–6) in the PVB, SAPB and PVB + SAPB groups, respectively ($P = 0.204$).

Except a medical history of COPD, no baseline characteristics was associated to the primary end-point (Supplementary Material, Fig. S1).

Secondary end-points

The groups did not differ significantly with regard to the VAS pain scores at any of the measurement time points (Figs 2 and 3). In all groups, the VAS pain score at rest ($P < 0.0001$) and on coughing fell between admission to the PACU and discharge from hospital ($P < 0.0001$). However, the decrease was significantly faster in the PVB + SAPB group than in the 2 other groups ($P = 0.0048$ for pain at rest and $P = 0.0037$ for pain on

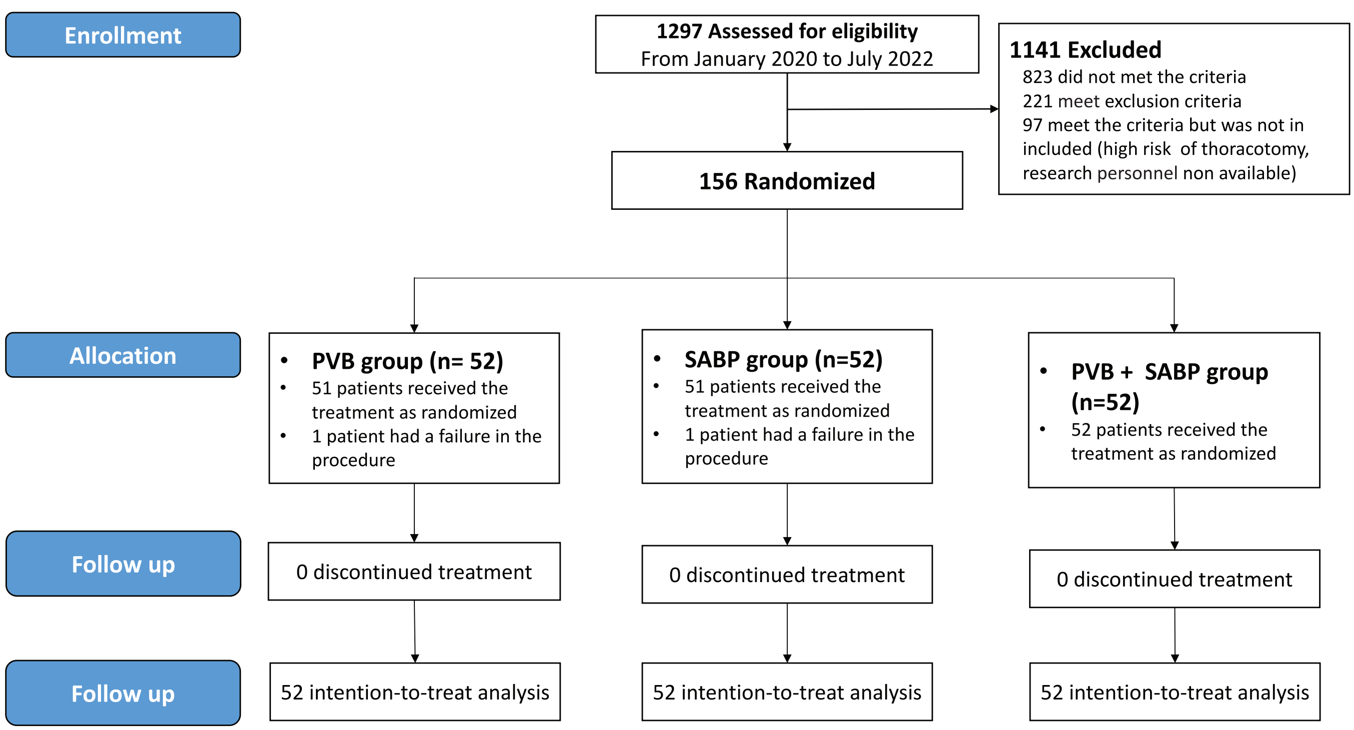


Figure 1: Study flow chart. PVB: paravertebral block; SAPB: serratus anterior plane block.

Table 1: Baseline and intraoperative data

Variables	PVB + placebo group ($n = 52$)	SAPB + placebo group ($n = 52$)	PVB + SAPB group ($n = 52$)	*P*-value
Age (years), median (IQR)	63 (56–71)	66 (60–71)	68 (59–73)	0.281
Male sex, *n* (%)	26 (50)	32 (62)	34 (66)	0.256
BMI (kg m^{-2}), median (IQR)	27.6 (23.3–31.9)	24.9 (21.6–28.9)	25.0 (22.8–27.9)	0.062
Resection type, *n* (%)				
Segmentectomy	6 (12)	8 (15)	4 (8)	0.103
Wedge	19 (37)	25 (48)	32 (61)	
Lobectomy	27 (52)	19 (37)	16 (31)	
Medical history, *n* (%)				
COPD	8 (15)	9 (18)	7 (14)	0.841
Current smoking	11 (21)	6 (12)	7 (14)	0.745
Previous smoking	23 (44)	26 (51)	25 (49)	
Diabetes	7 (14)	10 (20)	11 (21)	0.559
Coronary heart disease	2 (4)	7 (14)	7 (14)	0.181
Stroke	4 (8)	2 (4)	0 (0)	0.125
Depression	2 (4)	2 (4)	1 (2)	0.872
ASA physical status				
1	0 (0)	0 (0)	1 (2)	0.408
2	31 (60)	27 (52)	23 (44)	
3	41 (40)	25 (49)	27 (52)	
4	0 (0)	0 (0)	1 (2)	
Intraoperative period				
Ketamine use, *n* (%)	48 (92)	40 (80)	43 (83)	0.173
Operating time (min), median (IQR)	132 (88–174)	162 (106–198)	94 (68–162)	**0.005**
Conversion, *n* (%)	4 (8)	2 (4)	2 (4)	0.732
Revision surgery, *n* (%)	3 (6)	1 (2)	1 (2)	0.620
Prolonged air leakage, *n* (%)	4 (8)	4 (8)	7 (13)	0.656

The groups were compared in a chi-squared test, Fisher's exact test or the Kruskal–Wallis test.

$^*P < 0.05$.

ASA: American Society of Anaesthesiologists; BMI: body mass index; COPD: chronic obstructive pulmonary disease; IQR: interquartile range; PVB: paravertebral block; SAPB: serratus anterior plane block.

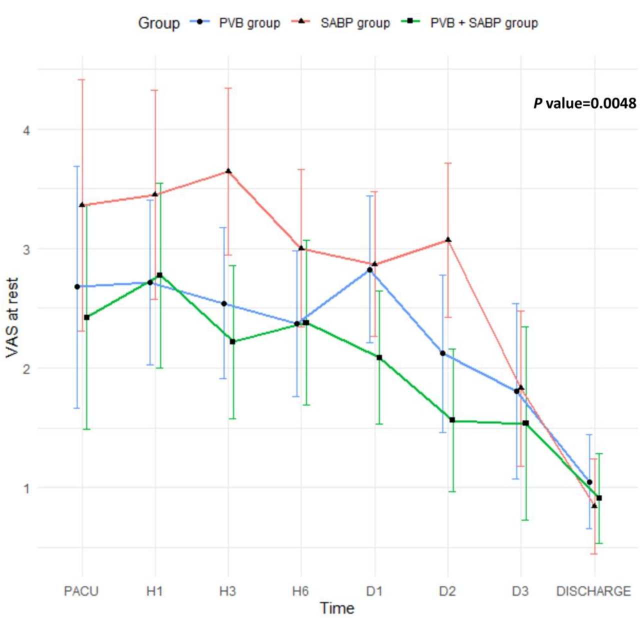


Figure 2: Pain intensity at rest, from admission to the postanaesthesia care unit (PACU) to hospital discharge. PVB: paravertebral block; SAPB: serratus anterior plane block; VAS: visual analogue scale.

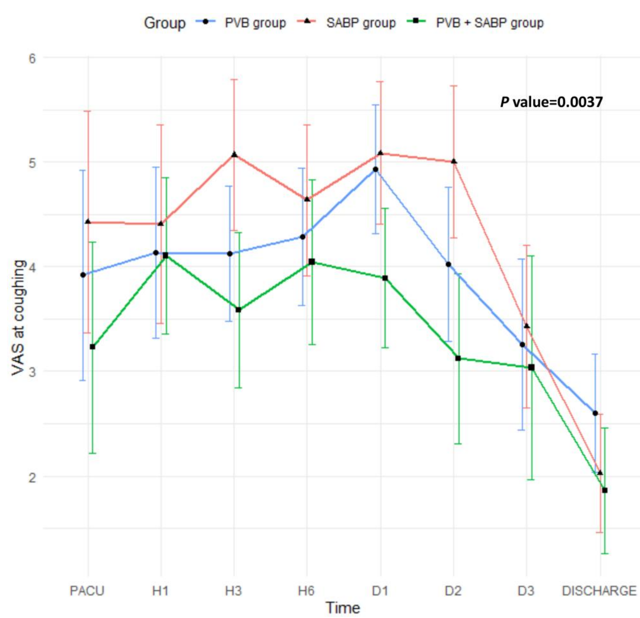


Figure 3: Pain intensity on coughing, from admission to the postanaesthesia care unit (PACU) to hospital discharge. PVB: paravertebral block; SAPB: serratus anterior plane block, VAS: visual analogue scale.

coughing). The 3 groups did not differ significantly with regard to cumulative opioid consumption [0 (0–14), 10 (0–25) and 3 (0–10) mg for the PVB, SAPB and PVB + SAPB groups, respectively; $P = 0.153$). Likewise, the groups did not differ significantly with regard to the hospital stay or the prevalence of pneumonia. Finally, neuropathic pain rate did significantly not differ between groups at month 6.

Adverse events

During the study enrolment and follow-up, no serious adverse event related to the regional anaesthesia procedure was notified.

DISCUSSION

Our study's main finding was that relative to PVB alone and SAPB alone, the combination of PVB and SAPB did not allow either a better pain relief right after surgery or a lower level of postoperative opioid consumption. However, the pain trajectory showed lower level of pain intensity on coughing and at rest from PACU admission to hospital discharge in the PVB + SAPB group. (Figs 2 and 3).

The present study is the 1st to have directly compared SAPB, PVB and the combination of the 2. To the best of our knowledge, 3 randomized trials have compared SAPB with PVB [17–19]. SAPB appeared to be non-inferior to PVB in terms of pain scores. However, the 3 trials had very different designs, which makes the results difficult to compare. First, one of the trials concerned thoracotomy, which is being supplanted by minimally invasive surgery [17]. Second, in one these trials, the authors used a catheter to ensure continuous infusion [18]. Third, and most importantly, the time point of pain assessment varied from 1 study to other (e.g. assessment on admission in one and 48 h after surgery in another). Nevertheless, our study confirmed the literature findings: SAPB and PVB have comparable effectiveness. Beyond regional anaesthesia, progress in surgery technique and pain management should be highlighted to demonstrate the superiority of one block to the other. First, it should be borne in mind that VATS is associated with lower postoperative pain levels than thoracotomy [20]. The relatively low pain score on admission to the PACU in all 3 study groups (Table 2) reflected the benefits of VATS. In addition, our surgical teams always strive to apply a conservative, muscle-sparing approach. In the light of our present results and the literature data, SAPB is probably not inferior to PVB. Second, all patients received the standard of care in multimodal pain management, i.e. systemic analgesia and regional anaesthesia [2]. Hence, the benefit of one type of block over the other is probably more difficult to demonstrate in a favourable context of multimodal pain management and VATS. Besides, the comparison of a block with a placebo alone would not be ethical, given the high level of evidence for the effectiveness of regional blocks. Finally, our surgical teams always strive to apply a conservative, muscle-sparing approach. In the light of our present results and the literature data, SAPB is probably not inferior to PVB.

Another key finding was that the PVB + SAPB combination was not associated with lower postoperative pain level at PACU admission when compared with the single blocks alone. To date, only 1 randomized trial has measured the efficacy of the PVB + SAPB combination [21]. In contrast to our study, Dusseaux *et al.* found that the PVB + SAPB combination was superior to PVB alone, with a significantly lower postoperative pain score 6 h after surgery. The cumulative dose of morphine was similar in the 2 groups. We used 40 mg of ropivacaine for the PVB but we used 80 mg the SABP, whereas Dusseaux *et al.* used 40 mg for the SAPB. When looking at pain over 24 h after surgery in Dusseaux *et al.*'s study, both groups (BVP and BVP + SAPB) have

Table 2: Primary and secondary end-points

Variables	PVB group (*n* = 52)	SAPB (*n* = 52)	PVB + SAPB group (*n* = 52)	*P*-value
Primary end-point				
VAS on coughing on admission to the PACU	3 (0–6)	4 (0–8)	2 (0–6)	0.204
Secondary end-points				
VAS on coughing after admission to the PACU				
Hour 1	4 (2–6)	5 (3–6)	4 (3–6)	0.0037
Hour 3	4 (3–5)	5 (3–7)	3 (2–5)	
Hour 6	4 (3–6)	5 (3–6)	4 (2–6)	
Day 1	5 (4–6)	5 (3–7)	4 (2–5)	
Day 2	3 (3–5)	5 (3–7)	3 (1–4)	
Day 3	3 (2–5)	3 (2–5)	3 (1–4)	
Hospital discharge	3 (2–4)	2 (0–3)	2 (0–2)	
VAS at rest after admission to the PACU				
Hour 0 (admission)	0 (0–5)	3 (0–6)	0 (0–5)	0.0048
Hour 1	3 (0–4)	3 (1–5)	2 (0–5)	
Hour 3	2 (1–3)	3 (2–5)	2 (0–4)	
Hour 6	2 (1–3)	3 (1–5)	2 (0–4)	
Day 1	3 (1–4)	3 (1–5)	2 (0–3)	
Day 2	2 (1–3)	3 (2–4)	1 (0–2)	
Day 3	2 (0–2)	1 (0–3)	1 (0–2)	
Hospital discharge	1 (0–2)	0 (0–2)	0 (0–2)	
Cumulative morphine use 72 h (mg)	0 (0–1)	10 (0–25)	3 (0–10)	0.153
Morphine titration in PACU, *n* (%)	2 (4)	1 (2)	1 (2)	0.649
Duration of chest drainage (days)	2 (1–3)	2 (1–3)	1 (1–3)	0.379
Length of hospital stay (days)	4 (3–6)	4 (3–6)	3 (2–5)	0.425
Pneumonia, *n* (%)	8 (15)	5 (10)	7 (13)	0.692
Neuropathic pain	9 (17)	5 (12)	5 (12)	0.545
DN4 at month 6	0 (0–2)	0 (0–1)	0 (0–0)	0.266

DN4: Douleur Neuropathique 4 questionnaire; NPSI: Neuropathic Pain Score Inventory; PACU: postanaesthesia care unit; PVB: paravertebral block; SAPB: serratus anterior plane block; VAS: visual analogue scale.

the same level suggesting that expected at hour 6, they did not how a benefit of the combination. In our study, the higher dose of 80 mg of ropivacaine in the SAPB could explain the lower pain intensity over hospital stay but that interpretation should be considered carefully as it was not our primary end-point [22].

Limitations

Our study had several limitations. As mentioned above, clinical studies of regional anaesthesia differed markedly in their design and methodology. First, we chose arbitrarily to assess the main end-point (pain) on admission to the PACU, although other time points could be more relevant. Indeed, the pain tended to be more intense 3 and 6 hours after PACU admission (Table 2). Another assessment time point might have highlighted a benefit of the PVB + SAPB combination. Second, the choice of our primary end-point could be challenged. Rather than selecting a single pain assessment time point, we could have selected the pain trajectory during the hospital stay as the primary end-point. In fact, the change over time in pain was a secondary end-point in our study. Wanderer and Rathmell described 5 distinct pain trajectories in the 7 days following surgery [23]. This approach offers a more dynamic assessment of pain and takes account of related subjective aspects (demographics characteristics, the patient's behaviour, previous experience of pain, etc.) [24]. Each patient becomes his/her own control, and the pain trajectory can be classified. In all 3 of our study groups, the patients' pain level decreased significantly between PACU admission and hospital discharge. However, the decrease was significantly faster in the PVB + SAPB group (Figs 2 and 3). Third, it is possible that some of our blocks failed. This is more of a problem for PVB with ultrasound guidance; this deep procedure can be challenging in patients with suboptimal echogenicity. In a study of a cohort of 620 adult patients, the PVB failure rate was around 6% [25]. We should probably have used a pinprick test at PACU admission to test the efficacy and level of the PVB after anaesthesia.

Concerning PVB, it is mentioned that very recently, a randomized trial confirmed the non-inferiority; a surgeon-performed *in situ* single shot in comparison to ultrasound-guided PVB by the anaesthetist [26]. That could be an answer to patient with poor echogenicity. Another alternative to manage pain could be the use of catheter for continuous infusion but it does not promote an early and easy patient mobilization. Finally, other less-invasive blocks such as erector spinae are well described and seems non-inferior to PVB, but the level of evidence remains uncertain.

Finally, the VATS techniques used in our institution (a modified three-port approach in the 7th intercostal space) might differ from those in other institutions; some surgical teams use an axillary port in the 4th or 5th intercostal plane or have access to robot-assisted thoracic surgery. The effectiveness of SAPB and PVB might depend on the type of surgery in general and the VATS technique in particular.

Despite these limitations, we believe that our double-blind, randomized trial provided robust data on the combination of SAPB and PVB in VATS.

Given our finding, the combination of PVB with SAPB is probably futile as they both target the intercostal nerve at 2 different sites. However, given the difficulty to proceed in PVB, further investigations should probably focus on non-inferiority of less-invasive blocks such as SAPB or erector spinae block in comparison to the PVB, which remains the gold standard. Although European Guidelines in Thoracic Surgery did not mention erector spinae or SAPB [1], a procedure-specific postoperative pain management recommended the use of SAPB or erector spinae block in VATS [2].

CONCLUSION

In the present study, PVB alone and SAPB alone provided similar levels of postoperative pain relief. We failed to demonstrate a benefit of the PVB + SAPB combination in the pain management, and so larger trials are now required.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *EJCTS* online.

ACKNOWLEDGEMENTS

Institution in which the work was performed: Department of Anesthesiology and Critical Care Medicine, Amiens University Hospital, Amiens, France.

FUNDING

This study was funded by Amiens Hospital University.

Conflict of interest: none declared.

DATA AVAILABILITY

The dataset used and analysed during the current study is available from the corresponding author upon reasonable request.

Author contributions

Florent Leviel: Conceptualization; Data curation; Writing—original draft; Writing—review and editing. **Alex fourdrain:** Conceptualization; Data curation; Writing—original draft; Writing—review and editing. **Florian Delatre:** Data curation; Formal analysis; Funding acquisition. **Florence De Dominicis:** Formal analysis; Writing—review and editing. **Thomas Lefebvre:** Data curation; Formal analysis. **Stéphane Bar:** Data curation; Formal analysis. **Hamza Yahia Alshatri:** Data curation. **Emmanuel Lorne:** Conceptualization; Data curation. **Olivier Georges:** Data curation; Formal analysis. **Pascal Berna:** Conceptualization; Data curation; Formal analysis. **Hervé Dupont:** Writing—review and editing. **Jonathan Meynier:** Formal analysis; Methodology. **Osama Abou-Arab:** Data curation; Formal analysis; Writing—original draft; Writing—review and editing.

Reviewer information

European Journal of Cardio-Thoracic Surgery thanks Mohsen Ibrahim, Luke A. Perry and the other, anonymous reviewer(s) for their contribution to the peer review process of this article.

REFERENCES

[1] Batchelor TJP, Rasburn NJ, Abdelnour-Berchtold E, Brunelli A, Cerfolio RJ, Gonzalez M *et al.* Guidelines for enhanced recovery after lung

surgery: recommendations of the Enhanced Recovery After Surgery (ERAS®) Society and the European Society of Thoracic Surgeons (ESTS). Eur J Cardiothorac Surg 2019;55:91–115.
[2] Feray S, Lubach J, Joshi GP, Bonnet F, Van De Velde M; the PROSPECT Working Group of the European Society of Regional Anaesthesia and Pain Therapy. PROSPECT guidelines for video-assisted thoracoscopic surgery: a systematic review and procedure-specific postoperative pain management recommendations. Anaesthesia 2022;77:311–25.
[3] Nair S, Gallagher H, Conlon N. Paravertebral blocks and novel alternatives. BJA Educ 2020;20:158–65.
[4] Baidya DK, Khanna P, Maitra S. Analgesic efficacy and safety of thoracic paravertebral and epidural analgesia for thoracic surgery: a systematic review and meta-analysis. Interact Cardiovasc Thorac Surg 2014;18:626–35.
[5] Yeung JHY, Gates S, Naidu BV, Wilson MJA, Gao Smith F. Paravertebral block versus thoracic epidural for patients undergoing thoracotomy. Cochrane Database Syst Rev 2016;2:CD009121.
[6] Tighe S, Greene MD, Rajadurai N. Paravertebral block. Continuing Education in Anaesthesia Critical Care & Pain 2010;10:133–7.
[7] Ökmen K, Metin Ökmen B. Evaluation of the effect of serratus anterior plane block for pain treatment after video-assisted thoracoscopic surgery. Anaesth Crit Care Pain Med 2018;37:349–53.
[8] Viti A, Bertoglio P, Zamperini M, Tubaro A, Menestrina N, Bonadiman S *et al.* Serratus plane block for video-assisted thoracoscopic surgery major lung resection: a randomized controlled trial. Interact CardioVasc Thorac Surg 2020;30:366–72.
[9] Toulouse E, Lafont B, Granier S, Mcgurk G, Bazin J-E. French legal approach to patient consent in clinical research. Anaesth Crit Care Pain Med 2020;39:883–5.
[10] Schulz KF, Altman DG, Moher D. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. J Pharmacol Pharmacother 2010;1:100–7.
[11] Hafiani E-M, Cassier P, Aho S, Albaladejo P, Beloeil H, Boudot E *et al.* Guidelines for clothing in the operating theatre, 2021. Anaesth Crit Care Pain Med 2022;41:101084.
[12] El-Boghdadly K, Wolmarans M, Stengel AD, Albrecht E, Chin KJ, Elsharkawy H *et al.* Standardizing nomenclature in regional anesthesia: an ASRA-ESRA Delphi consensus study of abdominal wall, paraspinal, and chest wall blocks. Reg Anesth Pain Med 2021;46:571–80.
[13] Plaud B, Baillard C, Bourgain J-L, Bouroche G, Desplanque L, Devys J-M *et al.* Guidelines on muscle relaxants and reversal in anaesthesia. Anaesth Crit Care Pain Med 2020;39:125–42.
[14] Fourdrain A, De Dominicis F, Iquille J, Lafitte S, Merlusca G, Witte-Pfister A *et al.* Intraoperative conversion during video-assisted thoracoscopy does not constitute a treatment failure†. Eur J Cardiothorac Surg 2019;55:660–5.
[15] Bouhassira D, Lantéri-Minet M, Attal N, Laurent B, Touboul C. Prevalence of chronic pain with neuropathic characteristics in the general population. Pain 2008;136:380–7.
[16] Attal N, Bouhassira D, Baron R. Diagnosis and assessment of neuropathic pain through questionnaires. Lancet Neurol 2018;17:456–66.
[17] Saad F, El Baradie S, Abdel Aliem MW, Ali M, Kotb TM. Ultrasound-guided serratus anterior plane block versus thoracic paravertebral block for perioperative analgesia in thoracotomy. Saudi J Anaesth 2018; 12:565–70.
[18] Hanley C, Wall T, Bukowska I, Redmond K, Eaton D, Ní Mhuircheartaigh R *et al.* Ultrasound-guided continuous deep serratus anterior plane block versus continuous thoracic paravertebral block for perioperative analgesia in videoscopic-assisted thoracic surgery. Eur J Pain 2020; 24:828–38.
[19] Wang Y, Shi M, Huang S, He X, Gu X, Ma Z. Ultrasound-guided serratus anterior plane block versus paravertebral block on postoperation analgesia and safety following the video-assisted thoracic surgery: a prospective, randomized, double-blinded non-inferiority clinical trial. Asian J Surg 2022;46:4215–21.
[20] Bendixen M, Jørgensen OD, Kronborg C, Andersen C, Licht PB. Postoperative pain and quality of life after lobectomy via video-assisted thoracoscopic surgery or anterolateral thoracotomy for early stage lung cancer: a randomised controlled trial. Lancet Oncol 2016;17:836–44.
[21] Dusseaux M-M, Grego V, Baste J-M, Besnier E, Boujibar F, Koscianski G *et al.* Paravertebral block combined with serratus anterior plane block after video-assisted thoracic surgery: a prospective randomized controlled trial. Eur J Cardiothorac Surg 2023;64:ezad170.
[22] Biswas A, Castanov V, Li Z, Perlas A, Kruisselbrink R, Agur A *et al.* Serratus plane block: a cadaveric study to evaluate optimal injectate spread. Reg Anesth Pain Med 2018;43:854–8.
[23] Wanderer JP, Rathmell JP. Cardiopulmonary bypass, renal oxygenation, & acute kidney injury. Anesthesiology 2017;126:A21.
[24] Vasilopoulos T, Wardhan R, Rashidi P, Fillingim RB, Wallace MR, Crispen PL *et al.*; Temporal Postoperative Pain Signatures (TEMPOS) Group. Patient and procedural determinants of postoperative pain trajectories. Anesthesiology 2021;134:421–34.
[25] Naja Z, Lönnqvist P-A. Somatic paravertebral nerve blockade. Incidence of failed block and complications. Anaesthesia 2001;56:1181–201.
[26] Chenesseau J, Fourdrain A, Pastene B, Charvet A, Rivory A, Baumstarck K *et al.* Effectiveness of surgeon-performed paravertebral block analgesia for minimally invasive thoracic surgery: a randomized clinical trial. JAMA Surg 2023;158:1255–63.